

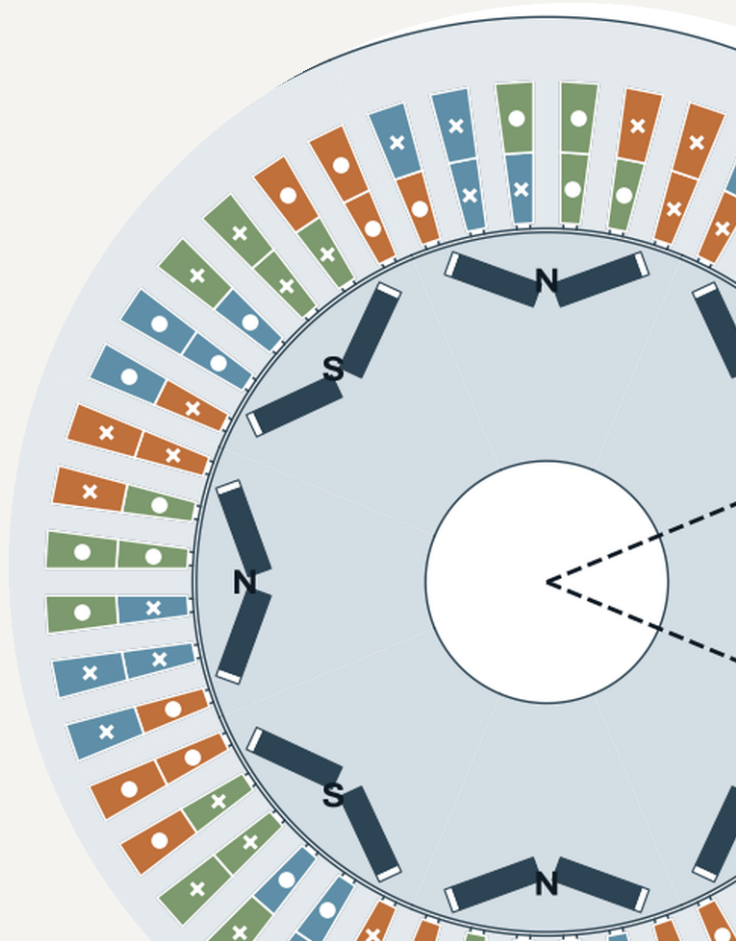
A DESIGN REFERENCE WITH A WORKED CASE

Designing an EV Traction Motor

From Vehicle Brief to
Finite-Element Validation

PREVIEW · DRAFT 1

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About this preview

This preview of *Designing an EV Traction Motor: From Vehicle Brief to Finite-Element Validation* holds the contents of the full book, the preface, the opening pages with the running case and the data labels, **Chapter 3** in full and the blank design card of **Appendix G**. The answers to the problems are in the full book.

The full book. About 290 pages in six parts and nine appendices: eighteen chapters with worked examples, derivations and problems with answers, the design card and public benchmark data. One running case, a B-segment range-extended EV with an 80 kW, 191 N·m traction motor, is carried from the vehicle brief to a finite-element check in FEMM. Every number carries its status: Confirmed, Inferred, Training target or Not available.

The companion tools. Six tools carry your numbers from one chapter to the next in a design card file: the Sizing Calculator, Topology Comparator, Materials Explorer, System Impact Calculator, Design Card Viewer and the FEMM Torque Lab. The Sizing Calculator of Chapter 3 is free at motoreka.my/tools.

The course. The book is the reference for Motoreka's EV Traction Motor course: two mornings in which engineering teams size, choose and check their own motor and defend it in a design review.

Get the book. The book is in preparation. To be told when it is out, to request an inspection copy for a course you teach, or to ask about the course for your team: motoreka.my/ev-traction-motor (Enquire).

Draft 1, October 2026. Numbers and wording may change before publication.

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Contents of the full book

The chapters in **bold** are in this preview. The full book has about 290 pages.

Part I · The brief and the rating

- 1 Why the motor matters
- 2 The design roadmap and the design card
- **3 From vehicle to motor torque and speed**

Part II · Sizing the motor

- 4 Main dimensions: from torque to rotor size
- 5 Electrical limits: poles, frequency, voltage and turns

Part III · Topology, materials and cost

- 6 How motors make torque
- 7 Choosing a topology
- 8 Components and electrical steel
- 9 Magnets
- 10 Windings and insulation
- 11 Cost of active materials

Part IV · The motor in the vehicle

- 12 The inverter and fault behaviour
- 13 Losses and cooling
- 14 Noise, vibration and mechanics

Part V · Finite-element validation

- 15 The one-pole finite-element model and the flux linkage
- 16 Torque, voltage limit, ripple and cogging
- 17 Closing the gap

Part VI · The design review

- 18 The design review

Appendices

- A Symbols and notation
- B The dq model, MTPA and field weakening

- C Materials and cost tables
- D Benchmark motors and vehicles
- E The FEMM Torque Lab: a guide
- F The companion tools and how to get them
- **G The blank design card and the review checklist**
- H Answers to selected problems
- I References and further reading

Preface

This book began with a problem that every motor designer meets sooner or later. A vehicle engineer arrives with a brief, a mass, a top speed, a battery voltage, and asks a plain question: *what motor do we need, and is it possible?* The answer is not one calculation. It is a chain of numbers, each of which limits the next, and the chain has to hold from the first estimate to the finite-element check.

I teach this chain in a two-morning training course, Motoreka's EV Traction Motor programme, for engineers who must work with motor designers without necessarily becoming one. The course is a good way to see the whole chain once. It is a poor reference. A participant goes home with slides and a spreadsheet, and a month later cannot remember why the shear stress was 50 kPa, where the 300 V came from, or what to change when the finite-element model disagrees with the hand estimate. This book is the reference that the two mornings could not be.

How it is written. One vehicle runs through every chapter: a B-segment range-extended EV with a peak of 80 kW and 191 N·m. Its numbers are carried from the brief to the final design card, and they change as the design is tightened. I have kept the changes visible rather than tidying them away. The hand estimate of the magnet flux linkage, 77.5 mWb, becomes 67.2 mWb in the finite-element model, and Chapters 15 to 17 spend their pages on why. A reader who sees a design go wrong and recover learns more than one who sees only the finished answer.

What I chose to leave out. The book does not derive the theory of electrical machines; excellent texts do that, and each chapter points to them. It gives public benchmark data, labels every number as confirmed, inferred, a training target or not available, and says which. Where I could not check a figure against a public source, I have said so.

How to read it. Straight through, if you are designing a motor: the chapters follow the order in which a design is done. By chapter, if you need one topic: each chapter begins with what it answers and ends with the lines it adds to the design card. With a calculator beside you in either case. The companion tools do the arithmetic once you have understood it, but every worked example is meant to be done by hand first.

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October 2026

Who this book is for

This book is for the engineer who has just been told: *design the traction motor for this vehicle*.

You may have come from the Motoreka training on EV traction motors, and kept this book as the reference that the two mornings could not be. You may have come from vehicle engineering, powertrain integration, or electrical engineering, and now sit across the table from a motor designer without yet speaking his language. Or you may be a motor engineer from another field, such as industrial drives, who now meets the constraints of a vehicle for the first time. In each case you need the same thing: a route from a vehicle brief to a motor that has been checked, in an order that makes sense, with numbers you can defend.

What the book assumes. You are comfortable with road-load equations, gear ratios and basic electrical circuits. You know what torque, power, voltage and current are, and you can use a spreadsheet or a short script. You do not need to have designed a motor before. The theory that you need is included, in boxed sections that you can skip on a first reading. One body of theory is placed in an appendix instead: the dq model, maximum torque per ampere and field weakening (Appendix B). It is useful from Chapter 5 on, and essential from Chapter 15.

What the book does not assume. It does not assume that you will compute everything by hand forever. The companion tools do the arithmetic once you have understood it. The book asks you to do each step by hand once, because a designer who has never done so cannot tell when a tool is wrong.

Who it is not for. It is not a research monograph. It does not derive the theory of electrical machines from first principles, survey the literature or present new results. Where a topic deserves more, each chapter ends with a short list of the books and papers that treat it properly.

What it solves

A new designer faces five problems in the first week.

1. **Where to start.** A motor has several hundred parameters. The book shows that most of them are fixed by a short chain of decisions that begins with the vehicle (Chapter 2).
2. **What order to work in.** Some decisions must come before others. The torque fixes the rotor size, the maximum speed fixes the pole count, the minimum bus voltage fixes the flux linkage. Taking them out of order leads to rework.
3. **What numbers are reasonable.** Every chapter gives typical values for the quantities it introduces, each with a label that says how far to trust it (see *Data labels* below).
4. **How to know you are right.** A first estimate from formulas is only a hypothesis. Part V shows how to test it with a finite-element model and how to read the gap between the model and the estimate.
5. **How to defend the result.** A design that cannot be explained to a vehicle team, a purchasing manager and a safety reviewer is not finished. The design card and the review of Chapter 18 are built for that.

The book follows one running case from the first page to the last, so that each chapter's numbers can be traced to the chapters before it.

What it does not solve. It does not replace detailed thermal, structural and acoustic analysis, a prototype and its tests, or the design of the controller. It sizes and checks the electromagnetic design of a permanent-magnet motor and shows how that design connects to the inverter, the cooling and the vehicle. Other machine types appear in the comparison of Chapter 7, but the worked design is an interior-magnet motor.

How to use this book

Three ways to read it.

Straight through. The chapters are written in the order in which a design is done. Parts I and II size the motor, Part III chooses its type and materials, Part IV places it in the vehicle, Part V validates it with finite elements, and Part VI is the design review.

As a reference. Each chapter stands on its own once you know the running case. Go to the chapter that matches your question, read its roadmap box to see where it sits, and use the box at the end, *Add to your design card*, to see what it commits you to.

Beside your own project. Keep two columns: the running case on the left and your own numbers on the right. Work through each step twice. The running case tells you whether your method is right; your own numbers tell you what your vehicle needs.

The elements of a chapter. Every chapter has the same parts, so that you can find them quickly.

Element	What it is for
Roadmap box	Shows where the chapter sits in the design chain
What you will learn	The outcomes you should be able to demonstrate afterwards
Theory and typical values	The method, with the ranges that real motors reach and a status label on each
<i>Derivation</i> boxes	Where a formula comes from. Skip on a first reading; return when you need to adapt it
<i>Worked example</i> boxes	The running case, step by step, with every number
<i>Companion tool</i> boxes	Which web tool or software repeats the calculation
<i>Common mistakes</i>	Errors that cost designers time
<i>Add to your design card</i>	The values that the chapter fixes
Problems and answers	Six problems; answers to selected ones are in Appendix H
Further reading	The sources to go to for more

The companion tools. Buyers of this book have access to the web tools of the training (the Sizing Calculator and the others) and to the FEMM Torque Lab software that is used in Part V. Appendix F explains what each tool does and how to obtain access. The book does not depend on them: every calculation can be done on a calculator, and the tools are for checking and for speed.

Conventions. Quantities are in SI units, except that speeds are given in rpm in the

text and in rad/s in equations. Torque is in N·m, stress in kPa, and flux linkage in mWb. Voltages and currents of the phase are peak values unless the text says rms. Angles of the dq model are electrical. Equations are numbered by chapter, such as (4.1), and so are figures and tables.

The running case

Every number in this book belongs to one vehicle, the *running case*: a B-segment range-extended EV (REEV). A REEV is driven only by its traction motor, through a single-speed reduction gear. A small engine and generator, and a battery, supply the DC bus. The designer's scope is the traction motor. The engine, generator, battery and the vehicle controller enter the problem only through their mass and the bus voltage.

The running case is generic. It is not the product of any manufacturer. Its numbers come from published sources, such as manufacturers' specifications for a comparable petrol car, drive-cycle regulations and published benchmark tests of production motors, and from estimates by EMDLab where nothing is published, with the method stated. The table below gathers its inputs.

Item	Value	Status
Base vehicle	B-segment petrol hatchback, kerb mass 1,025 kg	Confirmed
Range-extended conversion, kerb mass	1,225 kg (engine and gearbox removed, range extender, 20 kWh battery and motor added)	Inferred
Test mass / laden mass	1,300 kg / 1,600 kg	Inferred
Drag, frontal area, rolling resistance	$C_d = 0.296$; $A = 2.23 \text{ m}^2$; $C_{rr} = 0.011$	Inferred
Tyre rolling radius	0.283 m (185/55 R15)	Inferred
Performance targets	0–100 km/h in 9.5 s; 80–120 km/h in 7.0 s; 150 km/h top speed; 30 % hill start; 6 % climb at 80 km/h for 30 min; 30 s peak	Training target
DC bus	350 V nominal, 300 V minimum, 400 V maximum	Training target
Maximum motor speed	13,000 rpm	Training target
Packaging	Stator outer diameter at most 220 mm; active length at most 150 mm	Training target
Cooling	Water–glycol jacket, coolant inlet at most 65 °C; ambient 40 °C	Training target
Thermal limits	Winding hot spot at most 150 °C; magnets at most 140 °C	Training target

Inputs of the running case. All the targets and constraints were set for this book so that the design problem is well defined, and are typical of a B-segment EV drive.

Two real motors serve as benchmarks throughout: the 2012 Nissan Leaf motor (EM61), which is close to the running case in size, and a production B-segment EV, against which the book tests its sizing method. Appendix D collects their data and sources.

Data labels

A design book mixes numbers of very different reliability: a mass printed in a manufacturer’s brochure, a number estimated from a graph, a limit that the course set for convenience. A reader who cannot tell them apart will trust the wrong ones. So every table of values in this book carries one of four labels.

Label	Meaning	Use it as
Confirmed	Published by the manufacturer or measured in a cited test	A fact, to the precision published
Inferred	Estimated by EMDLab from published data, with the method stated	A good estimate; check it against your own data
Training target	Set for the running case so that the problem is well defined	A specification, not a measurement; replace it with your own
Not available	Not published; not used in the book	A gap you must fill from your own sources

One rule extends them. *A result calculated from a Training-target input is itself a Training target.* The rotor diameter of Chapter 4 follows from a peak torque that follows from targets set for the course, so it is a Training target, whatever the arithmetic. A result carries the weakest label among its inputs. This rule is stated in the tables as they occur, and the design card relies on it.

In your own project, replace the Training targets with the requirements of your customer and label your numbers in the same way. A design card with its sources and labels written down can be reviewed in an hour. One without them takes a week.

Symbols

The table below lists the symbols used most often. Appendix A has the full list, with the formulas, by chapter. Where a symbol has two meanings in common use, the book uses a different symbol for one of them (for instance, λ is the rotating-mass factor of Chapter 3, and λ_{pm} is the magnet flux linkage).

Symbol	Quantity	Unit
m_{test}, m_{lad}	Test mass, laden mass	kg
A, C_d, C_{rr}	Frontal area, drag coefficient, rolling-resistance coefficient	m ² , –, –
ρ, g	Air density, gravitational acceleration	kg/m ³ , m/s ²
r_w, G, η_g	Tyre rolling radius, reduction ratio, gear efficiency	m, –, –
λ	Rotating-mass factor	–
F_w, T_w	Tractive force and torque at the wheels	N, N·m
P_{pk}, P_{cont}	Peak and continuous shaft power	kW
T_{pk}, T_{cont}	Peak and continuous motor torque	N·m
$n_b, n_{max}, \omega_b, \omega_m$	Base speed, maximum speed (rpm); mechanical angular speed (rad/s)	rpm, rad/s
CPSR	Constant-power speed ratio n_{max}/n_b	–
D_r, L, D_{so}	Rotor diameter, active stack length, stator outer diameter	m or mm
σ	Air-gap shear stress	kPa
$\hat{B}_g, A$	Peak air-gap flux density, electric loading (rms)	T, A/m
k_w	Winding factor	–
p	Pole pairs (poles = $2p$)	–
f, ω_e	Electrical frequency, electrical angular speed ($= p\omega_m$)	Hz, rad/s
τ_p, Φ_p	Pole pitch, flux per pole	mm, mWb
$V_{dc}, \hat{V}_{ph}$	DC bus voltage, available peak phase voltage	V
λ_{pm}	Magnet flux linkage per phase (peak)	mWb
$\hat{E}_{LL}$	Peak line-to-line back-EMF	V
N	Series turns per phase	–

Symbol	Quantity	Unit
L_d, L_q, i_d, i_q	d-axis and q-axis inductance and current	mH, A

PART I

The brief and the rating

From the vehicle brief to the torque and speed that the motor must deliver. Chapters 1 to 3.

From vehicle to motor torque and speed

Brief → **Motor rating (this chapter)** → Torque sets rotor volume (Chapter 4) → Electrical limits (Chapter 5) → Topology, magnet, steel, winding (Chapters 6–11) → Finite-element validation (Chapters 15–17)

Every motor design starts from a vehicle, not from a motor. This chapter takes the performance targets of the vehicle and works backwards, step by step, to the five numbers that define the motor's rating: peak torque and power, continuous torque and power, base speed, maximum speed, and the ratio of the reduction gear. After this chapter you can write these numbers on the design card and every later chapter will build on them.

The method is the same for any vehicle. Only the inputs change. We use them here for the running case of this book: a B-segment range-extended EV, with a test mass of 1,300 kg and a laden mass of 1,600 kg.

IN THIS CHAPTER YOU WILL LEARN TO

- turn a set of vehicle targets into the force the wheels must deliver, using the road-load equation;
- say which targets set the **peak** rating and which set the **continuous** rating, and why;
- estimate the peak power for a given 0–100 km/h time with a hand formula, and check it with a time-stepped simulation;
- choose a reduction ratio, a base speed and a peak torque, and test them against hill start and wheel slip;
- test your whole method on a real production car.

You need: the vehicle brief: masses, drag, rolling resistance, tyre radius, and the six performance targets.

3.1 What the brief gives you

A traction motor is never asked to “make 80 kW”. It is asked to make a car accelerate, hold a speed on a hill, and start on a steep slope, and the 80 kW follows from those demands. The brief for the running case states six performance targets (Table 3.1). All six are Training targets: they were set for this book so that the design problem is well defined.

Table 3.1. Performance targets for the running case. All are Training targets.

ID	Target	Value	Condition
T1	Acceleration	0–100 km/h in 9.5 s or less	Test mass, nominal bus voltage
T2	Overtaking	80–120 km/h in 7.0 s or less	Test mass, nominal bus voltage
T3	Top speed	150 km/h or more	Level road, test mass, held continuously
T4	Hill start	30 % grade	Laden mass, standing start
T5	Sustained climb	6 % grade at 80 km/h	Laden mass, for 30 minutes
T6	Peak duration	30 s or more	Peak torque and power from a hot start

Each target belongs to one of two families, and the distinction runs through the whole book.

- **Peak duty** lasts seconds: accelerating, overtaking, starting on a hill. The motor and its inverter can exceed what the cooling system could carry for hours, because the windings and magnets heat up only a few degrees in that time. T1, T2 and T4 set the peak rating, and T6 says how long the peak must last.
- **Continuous duty** lasts as long as the driver wants: holding 150 km/h, or climbing a long hill. The motor must carry it indefinitely, with the heat balanced by the cooling. T3 and T5 set the continuous rating.

Besides the targets the brief gives constraints that the motor design must respect: the DC bus voltage (350 V nominal, 300 V minimum, 400 V maximum), the maximum motor speed (13,000 rpm), the packaging envelope (stator outer diameter at most 220 mm, active length at most 150 mm), the cooling (water-glycol jacket, inlet at most 65 °C) and the temperature limits (winding hot spot 150 °C, magnets 140 °C). Chapters 4 to 13 use them one by one. Here only the vehicle parameters of Table 3.2 matter.

Table 3.2. Vehicle parameters of the running case. For your own vehicle, replace these with your own data and record where each one came from.

Parameter	Symbol	Value	Basis	Status
Test mass	m_{test}	1,300 kg	Kerb mass plus a 75 kg driver	Inferred
Laden mass	m_{lad}	1,600 kg	Kerb mass plus five occupants and luggage	Inferred
Frontal area	A	2.23 m ²	0.85 × width × height	Inferred
Drag coefficient	C_d	0.296	Published claim for the base car	Inferred
Rolling resistance coefficient	C_{rr}	0.011	Passenger-car tyre on asphalt	Inferred
Air density	ρ	1.20 kg/m ³	About 30 °C at sea level	Inferred
Tyre rolling radius	r_w	0.283 m	Free radius 292 mm × 0.97 for load deflection	Inferred
Rotating-mass factor	λ	1.05	Single-speed EV driveline	Inferred
Reduction-gear efficiency	η_g	0.97	Single-stage helical gear	Training target

Notice the drag coefficient. A published claim and a measured value can differ a lot: a wind-tunnel study of an earlier model of the same class gave about 0.35, against the 0.296 claimed. A sound habit is to run the whole chain twice, once with each value, and see how much the answer moves (Problem 3.1).

3.2 The force at the wheels: the road-load equation

A vehicle moving at speed v on a road of slope angle α must overcome three resisting forces. The tractive force F_w that the wheels must deliver to hold the speed is their sum (Derivation 3.1):

$$F_w = mg \sin \alpha + C_{rr} mg \cos \alpha + \frac{1}{2} \rho C_d A v^2 \quad (3.1)$$

The three terms are the grade force, the rolling resistance and the aerodynamic drag. When the vehicle must also accelerate, a fourth term is added: $\lambda m dv/dt$, where λ accounts for the rotational inertia of the motor, gear and wheels.

The power the motor shaft must deliver is the tractive power divided by the gear efficiency,

$$P_m = \frac{F_w v}{\eta_g} \quad (3.2)$$

DERIVATION 3.1 THE ROAD-LOAD EQUATION

Take the vehicle as a point mass m on a straight slope of angle α , moving at a steady speed v . Newton's second law along the slope says the sum of the forces is zero, so the tractive force F_w equals the sum of the three resisting forces.

Grade. The component of weight along the slope is $mg \sin \alpha$. On level road $\alpha = 0$ and this term vanishes. A road grade is quoted as a percentage, which is the slope $\tan \alpha$, so a 6 % grade means $\alpha = \arctan 0.06 = 3.43^\circ$ and a 30 % grade means $\alpha = 16.70^\circ$.

Rolling resistance. The tyre deforms as it rolls, and the resisting force is proportional to the force pressing the tyre on the road, which is the component of weight perpendicular to the slope, $mg \cos \alpha$. The constant of proportionality is the rolling resistance coefficient C_{rr} , so $F_{roll} = C_{rr} mg \cos \alpha$.

Aerodynamic drag. A body of frontal area A moving through air of density ρ meets a dynamic pressure $\frac{1}{2} \rho v^2$. The drag force is this pressure times the frontal area times a shape factor, the drag coefficient C_d : $F_{aero} = \frac{1}{2} \rho C_d A v^2$. It grows with the square of speed, so it dominates at motorway speed.

Shaft power. Power at the wheels is $F_w v$. The reduction gear loses a fraction of what passes through it, so the motor must supply $F_w v / \eta_g$ when it is driving. (When the car brakes and the motor generates, the gear loss works the other way: the electrical side receives η_g times the wheel power.)

Acceleration. If the speed changes, the net force $F_w - F_{resist}$ accelerates the mass. The rotating parts must also be accelerated, and they contribute extra inertia equivalent to a fraction of the vehicle mass. Writing this as an effective mass λm with $\lambda \approx 1.05$ gives $F_w - F_{resist} = \lambda m dv/dt$.

Figure 3.1 shows what the equation gives for the running case. On the level road at test mass, rolling resistance is nearly constant and the aerodynamic force rises with v^2 . The motor power rises faster still, because power is force times speed: the aerodynamic part follows v^3 .

3.3 The duty points: what each target asks of the motor

With equations (3.1) and (3.2) the continuous targets can be evaluated directly. Each calculation is a **duty point**: a pair of speed and torque (or power) that the motor must be able to deliver for as long as the target demands.

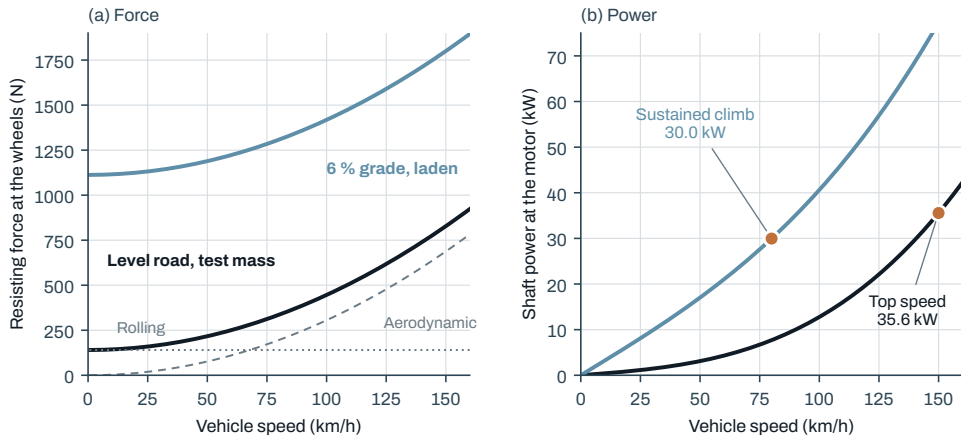


Figure 3.1. Road load of the running case. (a) The resisting force at the wheels against vehicle speed, on level road at test mass and on a 6 % grade at laden mass. (b) The shaft power the motor must supply to hold each speed, with the duty points of T3 and T5 marked.

WORKED EXAMPLE 3.1 THE CONTINUOUS DUTY POINTS

Top speed, T3. At 150 km/h, $v = 41.67$ m/s. On level road at test mass,

$$F_{aero} = \frac{1}{2}(1.20)(0.296)(2.23)(41.67)^2 = 687.6 \text{ N}, \quad F_{roll} = 0.011 \times 1300 \times 9.81 =$$

The shaft power is

$$P_{T3} = \frac{(687.6 + 140.3) \times 41.67}{0.97} = 35.56 \text{ kW}.$$

Sustained climb, T5. At 80 km/h, $v = 22.22$ m/s, with the laden mass 1,600 kg on a 6 % grade ($\alpha = 3.43^\circ$):

$$F_{grade} = 1600 \times 9.81 \times \sin \alpha = 940.1 \text{ N}, \quad F_{roll} = 172.3 \text{ N}, \quad F_{aero} = 195.6 \text{ N}.$$

$$P_{T5} = \frac{(940.1 + 172.3 + 195.6) \times 22.22}{0.97} = 29.97 \text{ kW}.$$

Continuous power. Take the larger of the two, add 10 % margin and round up to the next 5 kW:

$$P_{cont} = \lceil 1.10 \times 35.56 \rceil_5 = \lceil 39.1 \rceil_5 = 40 \text{ kW.}$$

Top speed, not the climb, sets the continuous rating of the running case.

Now the peak duties. Hill start is a torque problem, not a power problem. At standstill there is no aerodynamic drag, and the force is the grade plus the rolling resistance. For the laden car on a 30 % grade, $F_w = 1600 \times 9.81 \times (\sin 16.70^\circ + 0.011 \cos 16.70^\circ) = 4,676 \text{ N}$, and the wheel torque is $T_w = F_w r_w = 4,676 \times 0.283 = 1,323 \text{ N}\cdot\text{m}$. We will convert it to motor torque in the next section.

Acceleration is different again, because the motor power is needed to *increase the kinetic energy* of the car. For T1 there is a short hand formula (Derivation 3.2):

$$P_w \approx \frac{\lambda m (V_f^2 + V_b^2)}{2 t_a} + \frac{2}{3} mg C_{rr} V_f + \frac{1}{5} \rho C_d A V_f^3 \quad (3.3)$$

where t_a is the target time, V_f the final speed (100 km/h), and V_b the vehicle speed at which the motor reaches its base speed. The motor power is $P_m = P_w / \eta_g$. The first term dominates: accelerating a 1.3-tonne car to 100 km/h stores 527 kJ of kinetic energy (with λ), and the resistances add little.

DERIVATION 3.2 PEAK POWER FOR A GIVEN ACCELERATION TIME

A traction motor delivers constant torque up to its base speed and constant power above it. Neglect the road load for a moment, and let the motor power at the end of the constant-torque phase be P , reached at vehicle speed V_b .

Phase 1, constant force. Force $F = P/V_b$ is constant, so the speed rises linearly and the power rises linearly from 0 to P . The average power is $P/2$. The kinetic energy gained is $\frac{1}{2} \lambda m V_b^2$, so the phase takes

$$t_1 = \frac{\frac{1}{2} \lambda m V_b^2}{P/2} = \frac{\lambda m V_b^2}{P}.$$

Phase 2, constant power. The motor supplies P until the speed reaches V_f . The energy gained is $\frac{1}{2} \lambda m (V_f^2 - V_b^2)$, so

$$t_2 = \frac{\lambda m (V_f^2 - V_b^2)}{2P}.$$

Total. Adding,

$$t_a = t_1 + t_2 = \frac{\lambda m}{2P} (2V_b^2 + V_f^2 - V_b^2) = \frac{\lambda m (V_f^2 + V_b^2)}{2P},$$

which rearranges to the first term of equation (3.3).

The road load. The rolling force is nearly constant, so its energy is $C_{rr}mg$ times the distance travelled, which is the mean speed times t_a . In constant-power acceleration the speed grows as $v = V_f \sqrt{t/t_a}$, whose time-average is $\frac{2}{3}V_f$. This gives the second term. The aerodynamic power is $\frac{1}{2}\rho C_d A v^3$; the time-average of v^3 is $V_f^3 \cdot \frac{2}{5}$, so the average aerodynamic power is $\frac{1}{5}\rho C_d A V_f^3$, the third term.

The two averages are exact for a constant-power run and are used here for the whole run. This is a **hand estimate**: it ignores the limit on motor torque at low speed and the finite time spent at base speed. Section 3.5 checks it with a full simulation.

The base-speed value V_b depends on the reduction ratio and on the base speed, which we have not chosen yet. For the first pass use $V_b = 47$ km/h (13.1 m/s) and check it afterwards. With $t_a = 9.5$ s:

$$P_w = \underbrace{\frac{1.05 \times 1300 (27.78^2 + 13.06^2)}{2 \times 9.5}}_{67.7 \text{ kW}} + \underbrace{2.6}_{\text{roll}} + \underbrace{3.4}_{\text{aero}} = 73.7 \text{ kW}, \quad P_m = \frac{73.7}{0.97} = 76.0 \text{ kW}$$

So acceleration asks for about 76 kW at the shaft. The brief rounds this up to the **peak power of 80 kW**, which leaves about 5 % margin for the effects the hand estimate ignores. A peak-to-continuous ratio of $80/40 = 2$ is typical for a water-jacket-cooled traction motor.

3.4 From the wheel to the motor: reduction ratio, base speed and torque

A single-speed reduction gear of ratio G links the motor to the wheel. Speeds multiply by G and torques divide by $G\eta_g$:

$$n_m = G \frac{v}{r_w} \frac{60}{2\pi}, \quad T_m = \frac{T_w}{G\eta_g} \quad (3.4)$$

The ratio is the first real design choice, and it has a ceiling and a price.

The ceiling. At top speed the motor must not exceed its maximum speed $n_{max} = 13,000$ rpm. Setting $n_m = n_{max}$ at $v = 150$ km/h in equation (3.4) gives

$$G_{max} = \frac{\omega_{max} r_w}{v_{max}} = \frac{(13,000 \times 2\pi/60) \times 0.283}{41.67} = 9.25.$$

The price. A higher ratio means a smaller motor torque for the same wheel torque, so a smaller rotor (Chapter 4). It also means a higher motor speed at every vehicle speed, so a higher electrical frequency, more iron loss and more rotor stress (Chapter 5). The best choice sits near the ceiling, where the motor is small and still inside its speed limit. The brief takes $G = 9.0$, which puts the motor at 12,654 rpm at 150 km/h.

The hill-start check. With $G = 9.0$, the wheel torque of T4 becomes a motor torque of

$$T_m = \frac{1,323}{9.0 \times 0.97} = 151.6 \text{ N}\cdot\text{m}.$$

The peak torque of the motor must exceed it.

Base speed and peak torque. A traction motor delivers its peak torque up to the **base speed** n_b and its peak power above. The ratio n_{max}/n_b is the **constant-power speed ratio**, CPSR. A permanent-magnet motor with interior magnets typically achieves about 3 to 4 (Chapters 5 and 16 explain what limits it); the brief takes 3.25. Then

$$n_b = \frac{n_{max}}{\text{CPSR}} = \frac{13,000}{3.25} = 4,000 \text{ rpm}, \quad \omega_b = 418.9 \text{ rad/s},$$

$$T_{pk} = \frac{P_{pk}}{\omega_b} = \frac{80,000}{418.9} = 191.0 \text{ N}\cdot\text{m}, \quad T_{cont} = \frac{P_{cont}}{\omega_b} = \frac{40,000}{418.9} = 95.5 \text{ N}\cdot\text{m} \quad (3.5)$$

The vehicle speed at base speed is $V_b = \omega_b r_w / G = 418.9 \times 0.283 / 9.0 = 13.2 \text{ m/s} = 47.4 \text{ km/h}$, which agrees with the 47 km/h assumed in the hand estimate. If it had not, you would repeat the estimate with the new value.

Peak torque 191 N·m is above the hill-start torque 151.6 N·m, so T4 is met with a margin of 26 %. These numbers define the torque–speed envelope of Figure 3.2: constant torque up to 4,000 rpm, constant power from 4,000 to 13,000 rpm, for both the peak and the continuous ratings. The duty points of T3, T4 and T5 lie inside it.

The traction check. A motor that can deliver more torque than the tyres can transmit is wasted. With a friction coefficient $\mu = 0.9$ and 60 % of the test mass on the driven front axle, the largest motor torque the tyres can carry is

$$T_{slip} = \frac{\mu (0.6 mg) r_w}{G \eta_g} = \frac{0.9 \times 0.6 \times 1300 \times 9.81 \times 0.283}{9.0 \times 0.97} = 223 \text{ N}\cdot\text{m}.$$

The peak torque of 191 N·m is below it, so the front wheels can use it on dry asphalt. (On a wet road μ is lower and the traction control limits the torque; Problem 3.5.)

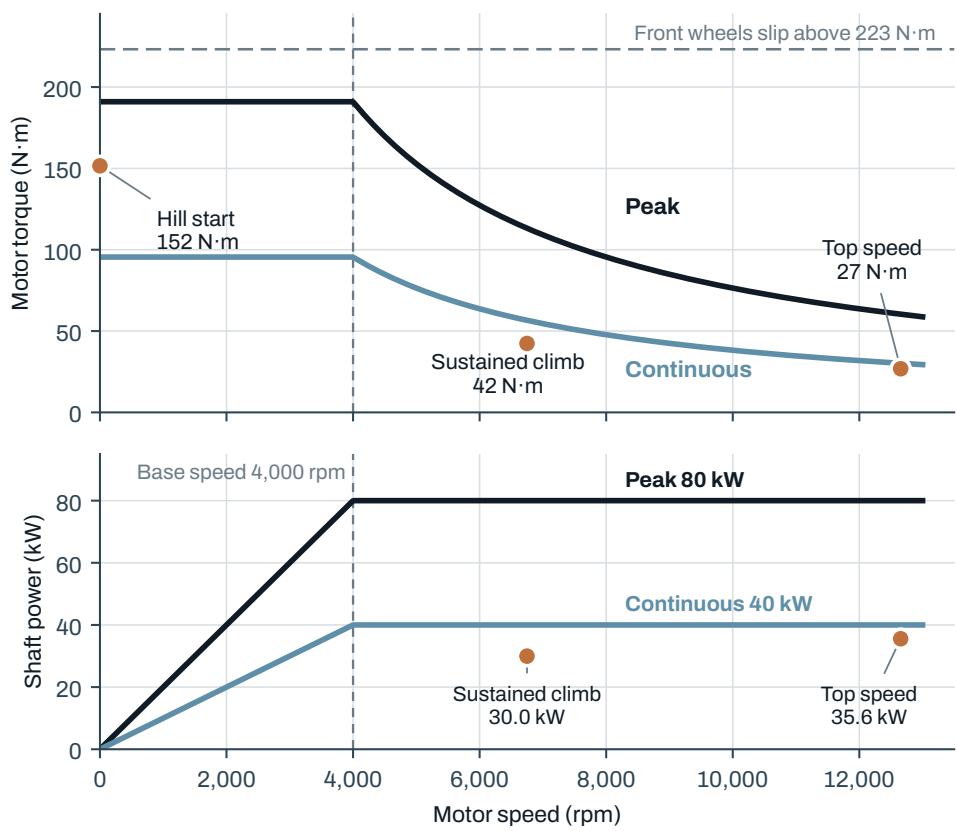


Figure 3.2. Torque and power envelopes of the running case, peak and continuous, with the duty points of the top-speed, climb and hill-start targets. Constant torque up to the base speed of 4,000 rpm; constant power above it. The front wheels slip above 223 N·m, well above the peak torque.

THE FIRST-PASS RATING IS NOT FINAL

Everything above is a first pass. It assumes a constant-power curve all the way to 13,000 rpm, which a real motor reaches only if its flux linkage and inductances are right (Chapters 5, 15 and 16). If the finite-element result shows a lower CPSR, the base speed rises, the peak torque at the same power falls, and the whole chain must be recomputed.

3.5 Checking the hand estimate with a simulation

Equation (3.3) is an estimate. To check it, step the vehicle through the full-throttle run in small time steps: at each step, find the motor speed from the vehicle speed, read

the available torque from the envelope of Figure 3.2, convert it to a force at the wheels, subtract the rolling and aerodynamic forces, and integrate the acceleration. The Sizing Calculator of the companion tools does exactly this. For the running case it gives

- 0–100 km/h in **8.95 s**, against the target of 9.5 s;
- 80–120 km/h in **6.55 s**, against 7.0 s;
- a minimum peak power of **75.8 kW** that satisfies both targets, against the 76.0 kW of the hand estimate.

The hand formula and the simulation agree to 0.3 %. The 80 kW of the design card therefore meets T1 and T2 with margin, and the margin is the 5 % that we added deliberately. Figure 3.3 shows the speed against time.

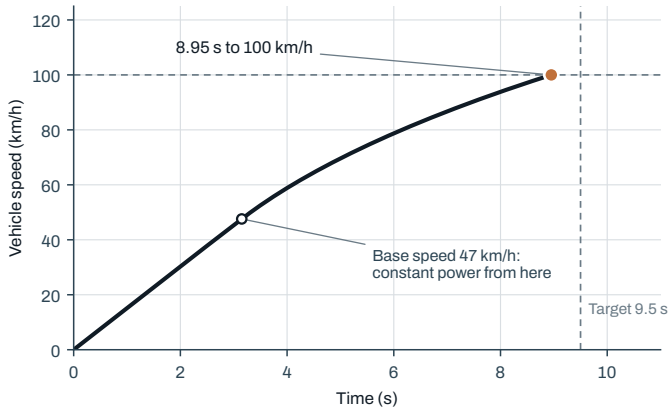


Figure 3.3. Simulated full-throttle run of the running case. The motor holds peak torque up to a vehicle speed of 47 km/h and peak power above it.

COMPANION TOOL

The Sizing Calculator runs the duty points, the envelope and the full-throttle simulation for your own vehicle data, and checks every target against the result. Do Worked Example 3.1 by hand first, then use the tool to check. The Sizing Calculator is free at motoreka.my/tools.

3.5.1 A reverse check on a real car

A method is only worth trusting if it reproduces a real vehicle. A production B-segment EV has public figures: a 150 kW motor, a 1,675 kg mass with driver, and a manufacturer-claimed 0–100 km/h time of 7.5 s. Rearrange equation (3.3) to find the time it predicts:

Group name
e.g. Group A

▼ Design choices

Peak power P_{pk} 80 kW

Continuous power P_{cont} 40 kW

Reduction ratio G 9 : 1

Constant-power speed ratio 3.25

Shear stress σ 50 kPa

Rotor diameter D_r 130 mm

Split ratio D_r/D_{st} 0.62

Poles 8

Peak gap flux density B_g 0.85 T

Winding factor k_w 0.933

$\cos \psi$ 0.9

Switching frequency 10 kHz

Back-EMF share at base 0.75

► Vehicle data (example car)

Checks ALL 10 CHECKS PASS

T1 0-100 km/h 8.95 s ≤ 9.5 s PASS	T2 80-120 km/h 6.55 s ≤ 7 s PASS	T3 Top speed held continuously 154 km/h at continuous power PASS
T4 Hill start 30 % 152 ≤ 191 N·m PASS	T5 Sustained climb 39.0 ≤ 40 kW continuous PASS	C2 Motor speed at top speed 12,654 ≤ 13,000 rpm PASS
C4 Envelope 210 mm OD, 145 mm long PASS	Tsp Rotor surface speed 88.5 ≤ 100 m/s PASS	f Switching per electrical cycle 11.5 ≥ 10 PASS
Grip Not traction limited 191 ≤ 223 N·m PASS		

MOTOREKA TRACTION MOTOR DESIGN CARD · SIZING

PEAK POWER	PEAK TORQUE	CONT. POWER	CONT. TORQUE	BASE SPEED	MAX SPEED
80 kW	191 N·m	40 kW	95.5 N·m	4,000 rpm	13,000 rpm
RATIO	ROTOR Ø × L	STATOR OD	SHEAR STRESS	POLES	MAX FREQUENCY
9.00 : 1	130 × 145 mm	210 mm	49.6 kPa	8	867 Hz

Torque and speed at the motor

Torque (N·m) Base 4,000 rpm n max

Figure 3.4. The Sizing Calculator, free at motoreka.my/tools, loaded with the running case. The panel at the left holds the design choices of Chapters 3 to 5. The ten checks and the design-card summary update as an input changes.

$$t_a = \frac{\lambda m (V_f^2 + V_b^2)}{2 (P \eta_g - P_{roll} - P_{aero})}.$$

That car's drag coefficient, frontal area and base speed are not all published, so assume $C_d = 0.30$, $A = 2.40 \text{ m}^2$, $V_b = 50 \text{ km/h}$, and keep $\eta_g = 0.97$. The rolling term is 3.35 kW and the aerodynamic term 3.70 kW, and the formula predicts **6.1 s**, against the claimed 7.5 s. The prediction is optimistic by about 1.4 s. That is the expected direction: the formula assumes the full peak power is available from the start, no wheel slip, and a motor at its rated power at the wheels. A real car has current limits in the battery and inverter at launch, a launch-control strategy that protects the drivetrain, tyre slip, and a power that may be rated at a higher bus voltage than the one the battery gives under load. The method therefore tells you what the powertrain *could* do, and a real car meets it only if all these effects are managed. This is why we leave margin on the peak power rather than sizing exactly to the estimate.

COMMON MISTAKES

- **Sizing the continuous rating from acceleration.** T1 and T2 last seconds; they set the peak rating. The continuous rating is set by the longest-duration duty, here T3 and T5.
- **Using the wrong mass.** The test mass (with driver) is for acceleration and top speed; the laden mass is for gradeability. Using the laden mass

for acceleration oversizes the peak power by about 20 %.

- **Forgetting the gear efficiency.** The motor must supply more than the wheels receive; dropping η_g underestimates the power by 3 %.
- **Converting grade percent as an angle.** A 30 % grade is $\arctan 0.30 = 16.7^\circ$, not 30° and not 0.30 rad. For small grades $\sin \alpha \approx \tan \alpha$, but at 30 % the difference is 4 %.
- **Choosing the ratio from top speed alone.** The ceiling G_{max} guarantees the speed limit. The hill-start torque and the traction limit must still be checked against the peak torque.
- **Taking the base speed from the target instead of the design.** V_b in equation (3.3) is a result of G and n_b . Check it after the ratio is set, and repeat the estimate if it moves.

3.6 Summary and design card

KEY EQUATIONS OF THIS CHAPTER

Tractive force and shaft power:

$$F_w = mg \sin \alpha + C_{rr} mg \cos \alpha + \frac{1}{2} \rho C_d A v^2, \quad P_m = F_w v / \eta_g$$

Peak power for an acceleration time t_a :

$$P_m \approx \frac{1}{\eta_g} \left[\frac{\lambda m (V_f^2 + V_b^2)}{2 t_a} + \frac{2}{3} m g C_{rr} V_f + \frac{1}{5} \rho C_d A V_f^3 \right]$$

Gear, base speed and torque:

$$G_{max} = \frac{\omega_{max} r_w}{v_{max}}, \quad n_b = \frac{n_{max}}{\text{CPSR}}, \quad T = \frac{P}{\omega_b}, \quad T_m = \frac{T_w}{G \eta_g}$$

Rules of the method: peak from acceleration and hill start, continuous from top speed and sustained climb; choose G just below G_{max} ; check the hill start and the traction limit against the peak torque; check the hand estimate by simulation.

ADD TO YOUR DESIGN CARD

Field	Running case
Peak power	80 kW
Peak torque	191.0 N·m
Continuous power	40 kW
Continuous torque	95.5 N·m
Base speed	4,000 rpm
Maximum speed	13,000 rpm
Reduction ratio	9.0

Every value on this page is a Training target, because it follows from targets that were set for this book. Chapter 4 uses the peak torque to size the rotor.

PROBLEMS

- 3.1** Repeat the top-speed duty point with $C_d = 0.35$ instead of 0.296. What is P_{T3} ? What continuous power would the rule “add 10 %, round up to the next 5 kW” then give? How much does the choice of drag coefficient move the continuous rating?
- 3.2** If the motor’s maximum speed were limited to 12,000 rpm instead of 13,000 rpm, what is the largest reduction ratio that keeps the speed limit at 150 km/h?
- 3.3** Raise the ratio to $G = 10$. What is the motor speed at 150 km/h? Is the 13,000 rpm limit still met? What is the hill-start torque at the motor?
- 3.4** A motor with a constant-power speed ratio of 4.0 instead of 3.25 is proposed, with the same 80 kW and 13,000 rpm. What are the base speed and the peak torque? By what fraction does the rotor volume at constant shear stress change relative to the running case (Chapter 4)?
- 3.5** What friction coefficient μ do the front wheels need to transmit the peak torque of 191 N·m at test mass, with 60 % of the mass on the front axle and $G = 9.0$? Is that realistic for a wet road?
- 3.6** Give three reasons why the 0–100 km/h time predicted by equation (3.3) for the production car is shorter than the claimed time.

Answers to selected problems are in Appendix H of the full book.

Further reading

- Ehsani, M., Gao, Y., Longo, S. and Ebrahimi, K., *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*, CRC Press, 3rd ed., 2018. The road-load equation, the acceleration-power approximation and the choice of the reduction ratio.
- Gillespie, T. D., *Fundamentals of Vehicle Dynamics*, SAE International, revised ed., 2021. Rolling resistance, drag and traction limits.
- The manufacturer's published specification and press material, for the figures of the reverse check (Appendix D of the full book lists the sources).

The blank design card and the review checklist

Copy the card for your own design. Fill in the header and the brief when you start (Chapter 2), add the values that each chapter fixes as you go, and use the checklist at the end before a review (Chapter 18). Give every number a source and a status: Confirmed, Inferred, Training target or Not available.

The design card

Table G.1. The blank design card. Status: not applicable (a template).

Field	Your design	Source and status
Header		
Project · case · date · version		
Designer · reviewer		
Brief and rating (Chapters 3, 5)		
Peak power · peak torque		
Continuous power · continuous torque		
Base speed · maximum speed		
Reduction ratio		
DC bus, minimum · maximum		
Acceleration targets (0–100, 80–120 km/h)		
Size (Chapter 4)		
Rotor diameter · stator outer diameter		
Stack length · shear stress		
Poles · slots · slots per pole per phase		
Electrical frequency at maximum speed		
Electrical targets (Chapter 5)		
Magnet flux linkage (cold · hot)		
Turns per phase · paths · conductors per slot		
$L_d \cdot L_q \cdot$ source		
Peak current · inverter apparent power		
Back-EMF at maximum speed		
Topology and materials (Chapters 7 to 11)		
Topology · alternative kept		
Steel grade · thickness		
Magnet grade · size · mass		
Winding type · pitch · strands		
Insulation class		
Active-material cost · date of prices		
System (Chapters 12 to 14)		
Power device · module class		
Active short-circuit current · crossover speed		

Field	Your design	Source and status
Thermal design case · hot spot		
Cogging order · radial mode		
Bridge safety factor · k_t source		
Finite-element check (Chapters 15 to 17)		
λ_{pm} (FE) · back-EMF (FE)		
Torque at MTPA · phase voltage at that point		
Torque inside the voltage limit		
Average torque · ripple · skewed or straight		
Proposed change and its cost		
Review (Chapter 18)		
Binding limit		
Margins that are assumed · unknown		
Decision · open risks · before prototype		

The review checklist

Tick each line, with the number of the card field that shows it.

Traceability

- () Every number has a source and a status label.
- () Peak and continuous values are kept apart.
- () A number from a Training-target input is itself a Training target.
- () Every price has a date.

Limits and consistency

- () Volume, frequency, voltage and thermal limits are all checked, and the binding one is named.
- () The voltage is checked at the same operating point as the torque.
- () λ_{pm} , L_d , L_q , turns and current agree with each other.
- () The hot and the cold values are not mixed (torque is delivered hot; fault voltage is a cold quantity).

Choices

- () Each choice has a trade-off stated with a number, and an alternative is kept.

- () The cost of every change is stated: magnet mass, inverter current, losses.

Finite-element evidence

- () The gap to the hand estimate is stated and explained.
- () The torque definition is stated.
- () What the model leaves out is listed, with its likely effect on the margin.

Margins and decision

- () Each margin is marked earned, assumed or unknown.
- () The decision is stated: go, go with conditions, or redo.
- () Each open risk has an owner, a closing analysis and a gate.

Continue in the full book

Chapter 3 gave the motor its rating: 80 kW and 191 N·m, a base speed of 4,000 rpm and a 9 : 1 gear. The rest of the book turns that rating into a motor:

- **Chapters 4 and 5** size the rotor from the shear stress, choose the poles and frequency, and fit the flux and the turns to the bus voltage.
- **Chapters 6 to 11** explain how the motor makes torque, choose the topology, the steel, the magnet grade and the winding, and cost the active materials.
- **Chapters 12 to 14** check what the motor asks of the inverter and the cooling: faults, losses, hot spots, noise and rotor stress.
- **Chapters 15 to 17** build one pole of the motor in FEMM, compare it with the hand estimate and close the gap.
- **Chapter 18** defends every number of the design card in a design review.

motoreka.my/ev-traction-motor: be told when the book is out, request an inspection copy, or bring the course to your team. The Sizing Calculator is free at **motoreka.my/tools**.